

## REVIEW

# Review on Image Preprocessing and Driving Waveform Design for Electrophoretic Displays

Xiangjie Zhong  | Xiaoyan Zhao | Tiesong Zhao

Fujian Key Laboratory for Intelligent Processing and Wireless Transmission of Media Information, College of Physics and Information Engineering, Fuzhou University, Fuzhou, Fujian, China

**Correspondence:** Tiesong Zhao ([t.zhao@fzu.edu.cn](mailto:t.zhao@fzu.edu.cn))

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## ABSTRACT

E-paper, as an emerging eco-friendly display technology, has gained significant attention due to its paper-like visual effect and low power consumption characteristics. The concept was first proposed in the 1970s, and after years of development, it has evolved from the initial black-and-white image display technology to full-color video display. Electrophoretic display shows images by controlling charged particles within microcapsules using an electric field, offering high reflectivity and contrast, and is widely used in various fields such as e-readers, mobile phones, automobiles, wearable devices, and more. The e-paper display system mainly consists of the e-paper screen, control circuits, image processing algorithms, and electronic ink driving waveforms. The review introduces the research history and applications of electrophoretic display, e-paper image processing algorithms, and driving waveforms. It also summarizes the current challenges and development trends in the field of e-paper.

## 1 | Introduction

As an emerging green and low-carbon display technology, electronic paper has broad application prospects and is receiving increasing attention. With the development and deployment of Internet of Things, as well as the promotion of eco-friendly and paperless initiatives, the e-paper industry is embracing a significant opportunity for growth. E-paper is a display device that changes its own color to allow viewers to perceive images through reflected light. In terms of visual performance and eye strain, e-paper resembles traditional paper [1]. Electrophoretic display (EPD) was developed in 1997 by J.D. Albert and Barrett Comiskey at the Massachusetts Institute of Technology [2]. Since the colored particles in electrophoretic display no longer require energy to maintain their position once they are driven to their display state, the technology is known for its low power consumption [3]—which is why it was initially used in e-readers.

Over years of development, e-paper has expanded its applications from e-readers [4, 5] to consumer products like smartphones,

automotive displays, and wearables [6] and further into fields such as healthcare, retail [7, 8], and transportation [9, 10]. As e-paper performance continues to improve, its range of applications is becoming increasingly broad. A system used for electrophoretic display consists of a processor and an EPD device, as shown in Figure 1. The processor includes an input interface, storage module, power module, control module, and e-paper driving module.

Before the system displays an image, the controller reads the external input, and the drive waveform lookup table (LUT) and the image data to be displayed are input into the storage module. The image has been processed by the EPD-specific enhancement algorithm before input. When displaying the image, the controller uses the EPD drive module to read the image to be displayed from the storage module, obtains the voltage sequence corresponding to the drive waveform LUT, and controls the power module to input the voltage sequence to the gate and source driver ICs. The EPD displays according to the voltage signal.

## 2 | Electrophoretic E-Paper and Development

E-paper display technologies include electrowetting display (EWD), electrochromic display (ECD), and EPD. The latter encompasses various architectures such as microcapsule, microcup, and liquid powder displays (LPDs), all operating on the principle of electrophoretic particle migration under an applied electric field [11]. Due to its high reflectivity and strong contrast, EPD currently holds a major share in the e-paper market [12]. The structure of electrophoretic e-paper consists of microchambers, charged particles, solvents, dispersants and stabilizers, external adhesive layers, and electrodes connected to the driving backplane [13]. The working principle of EPD is based on the movement of charged particles in a solvent under the influence of an electric field [14]. As shown in Figure 2, in a black-and-white microcapsule EPD, when the bottom electrode is negatively charged, white particles move to the top surface, displaying a white pixel. Conversely, when the electrode is positively charged, black particles move to the surface, resulting in a black pixel.

In the early stages of EPD development, Comiskey et al. [15] pioneered microencapsulated electrophoretic ink, enabling printable, bistable displays with extended service life—a foundational breakthrough that catalyzed the emergence of commercial e-paper. Hou et al. [16] proposed a novel EPD structure that achieves high contrast and fast response black and white image display in a transparent solution by using black and white particles with

opposite surface charges and controlled by an active matrix driving method. To facilitate scalable manufacturing, Zang et al. [17] introduced the microcup architecture and roll-to-roll fabrication process, significantly improving yield and uniformity for mass production. Concurrently, Henzen et al. [4] demonstrated the first active-matrix EPD integrated with thin-film transistor (TFT) backplanes, enabling high-resolution addressing and partial image updates essential for e-reader applications. As EPDs entered consumer markets, waveform design and image processing became critical for visual quality: Amundson et al. [18] established multi-level grayscale driving schemes.

Following the success of black-and-white displays, researchers turned to full-color EPDs. Kim et al. [19] proposed 300-micron color bars of cyan, magenta, yellow, and black microcapsules (Figure 3a). Subsequently, it was proposed to add a color filter array (CFA) on the basis of black-and-white EPDs [20, 21], as shown in Figure 3b. Lenssen et al. [22] suggested stacked microchambers with different colored particles, using electric fields to control visibility (Figure 3c). Mukherjee et al. [23] introduced a biprimary RGB-CMY system within single pixels (Figure 3d). E Ink's technology achieves full-color EPDs with high color saturation by filling microcups with three or four types of colored particles, such as cyan, magenta, yellow, and white charged pigment particles [24–27], as shown in Figure 4.

While these five approaches all aim to achieve full-color EPDs, they differ significantly in optical performance,

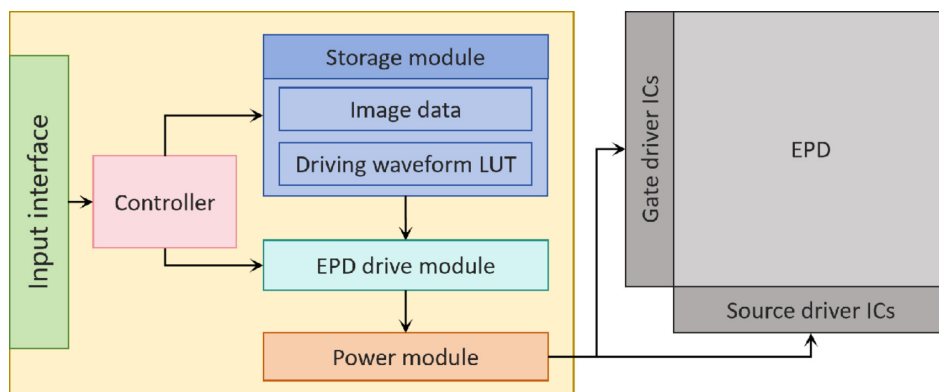


FIGURE 1 | Electrophoretic display system.

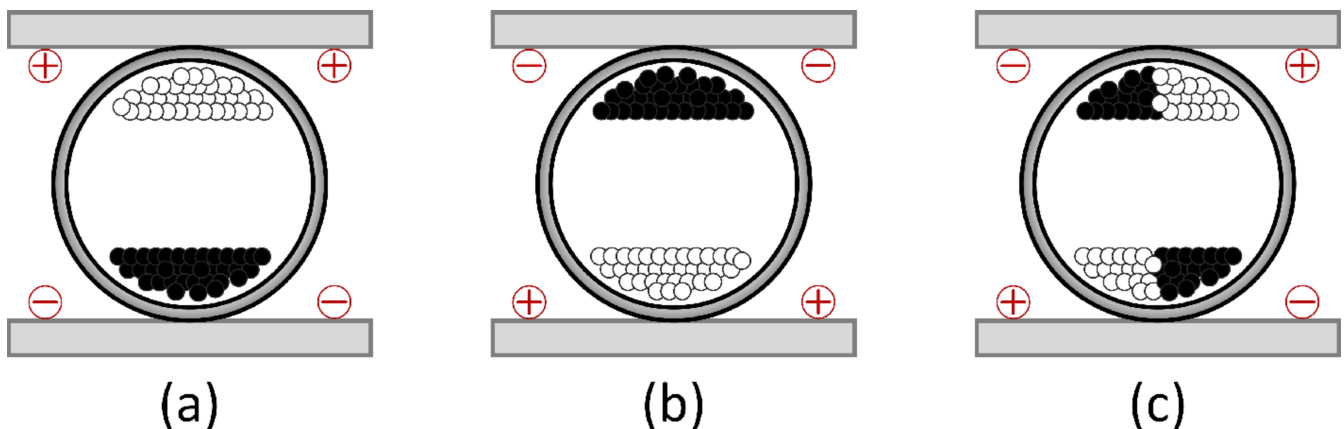
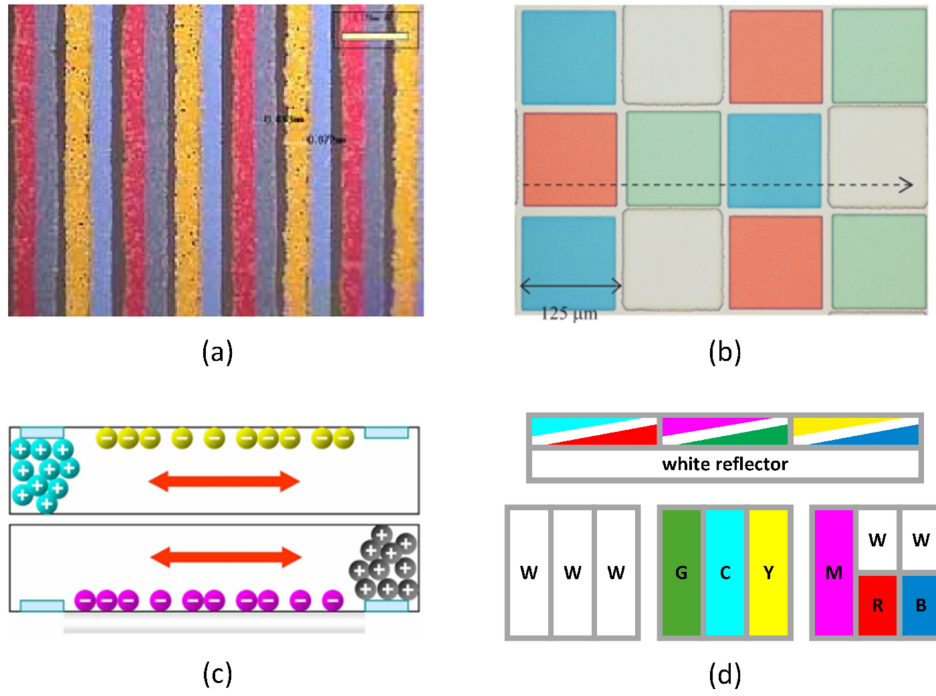
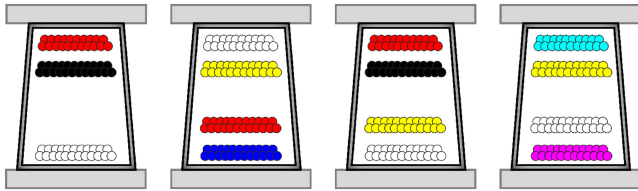


FIGURE 2 | Microcapsule electronic paper: (a) white; (b) black; (c) gray.



**FIGURE 3** | Schematic diagram of color electronic paper implementation: (a) color pixel bars [19]; (b) color filter [20]; (c) multilayer microcavity stacking [22]; (d) complementary dual primary colors [23].



**FIGURE 4** | Three-color and four-color micro cup electronic paper [24–27].

manufacturing complexity, and commercial viability. The CFA method offers the easiest integration with existing monochrome EPD production lines but suffers from low reflectance and narrow color gamut. In contrast, multi-pigment microcup and color bar methods provide higher brightness and saturation but face challenges in color mixing and resolution. Stacked microcavities and biprimary systems offer theoretical advantages in color gamut and pixel-level control but remain largely experimental due to fabrication and driving complexities. Among these, only the CFA and multi-pigment microcup approaches have achieved commercial deployment, reflecting the industry's prioritization of manufacturability and reliability over theoretical performance.

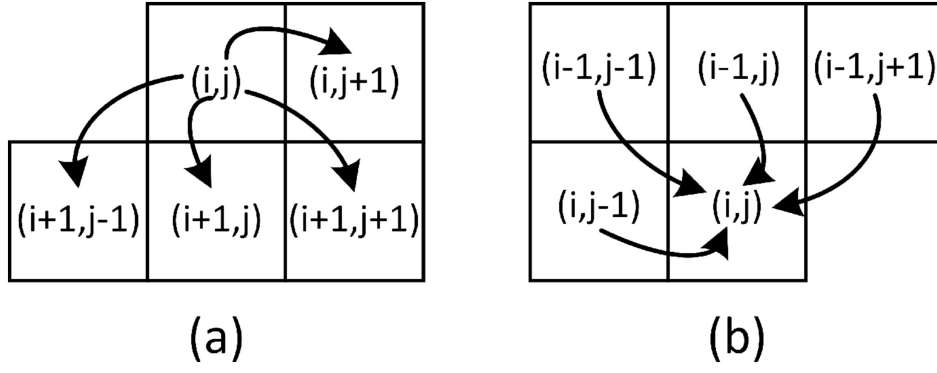
### 3 | Image Processing Algorithms

Due to the limited number of gray levels (2–16 discrete gray levels) in EPDs, it is necessary to convert continuous tone images into limited gray-scale patterns through digital halftoning techniques before display to improve the image quality perceived by the human eye. This is a technique originally developed for printing [28], which simulates gray levels or colors by spatially modulating the density of white/black dots within local

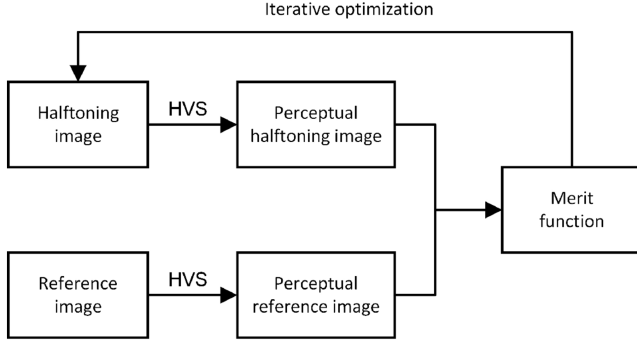
**TABLE 1** | Comparison of halftoning algorithms.

| Halftoning algorithm | Advantages                                                                                                | Disadvantages                                                                            |
|----------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Dithering            | The algorithm is simple and easy to implement.                                                            | Regular structural textures will appear.                                                 |
| Error diffusion      | 1. Rich tonal range in halftone image.<br>2. Anisotropic and irregular pixel distribution.                | 1. Moire phenomenon.<br>2. Highlight or dark areas may show lag in processing direction. |
| Iterative method     | 1. Excellent visual quality, no moire or artificial textures.<br>2. Accurate reproduction of image tones. | 1. Complex computation.<br>2. Difficult to apply in real-time processing.                |

neighborhoods. The three foundational halftoning approaches [29] are: (1) ordered dithering, which uses a fixed threshold matrix; (2) error diffusion, which propagates quantization error to neighboring pixels to preserve tone; and (3) iterative optimization (e.g., Direct Binary Search [DBS]), which minimizes perceptual error through repeated refinement. Table 1 summarizes their trade-offs in visual quality, computational cost, and artifact generation. While early EPD systems adopted basic variants



**FIGURE 5** | Principle of error diffusion algorithm: (a) error propagation; (b) error reception.



**FIGURE 6** | Schematic diagram of DBS algorithm [33].

of these methods, recent work has tailored them to address EPD-specific challenges such as ghosting, temperature sensitivity, and driving current constraints.

$$\mathbf{K} = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} \\ k_{21} & k_{22} & k_{23} & k_{24} \\ k_{31} & k_{32} & k_{33} & k_{34} \\ k_{41} & k_{42} & k_{43} & k_{44} \end{bmatrix} = \begin{bmatrix} 0 & 14 & 3 & 13 \\ 11 & 5 & 8 & 6 \\ 12 & 2 & 15 & 1 \\ 7 & 9 & 4 & 10 \end{bmatrix} \quad (1)$$

### 3.1 | Grayscale Display Processing Algorithms

For black-and-white image display, image processing algorithms are primarily used to improve grayscale contrast and enhance image detail. To optimize the visual performance of EPDs, Kao et al. [30] applied a dithering-based halftoning algorithm combined with contrast enhancement. The dithering method uses a Nasik matrix (as shown in Equation 1) to convert each pixel of a color image into a 16-level grayscale black-and-white binary grid. Among ordered dithering methods, the Nasik square is often used in EPDs to generate 16-level grayscale from a  $4 \times 4$  binary pattern. As shown in Equation (1), the matrix  $K$  contains integers from 0 to 15; for an input pixel with normalized intensity  $I \in [0, 15]$ , the output is white if  $I > K_{ij}$ , black otherwise. This yields 16 distinct dot densities, enabling smooth tone reproduction without Moiré patterns.

The error diffusion algorithm processes pixels in a raster scan order. For each pixel at  $(i,j)$ , the quantization error—the

difference between its original grayscale value and the binarized output (0 or 255)—is distributed to four neighboring pixels that have not yet been processed (right, bottom-left, bottom, and bottom-right) with weights  $7/16$ ,  $3/16$ ,  $5/16$ , and  $1/16$ , respectively. This is error propagation (Figure 5a). Conversely, each pixel receives accumulated errors from four previously processed neighbors (left, top-left, top, and top-right) before its own thresholding (Figure 5b). This bidirectional error flow preserves the local average luminance, as the weighted contributions ensure that the total error within any local region sums to zero over the entire image [31]. Feng et al. [32] proposed a history-aware error diffusion halftoning algorithm that incorporates information from the previous frame during the halftoning of the current image. Specifically, they modeled the display error—defined as the reflectance deviation caused by pixel state transitions—as a LUT. During the error diffusion process, this display error is fed back and combined with the conventional quantization error for diffusion. By constructing a display error LUT and integrating it into the halftoning feedback loop, their method significantly reduces ghosting in EPDs without increasing hardware cost, enabling high-quality, low-latency image updates.

The DBS algorithm is a representative iterative halftoning technique. As shown in Figure 6, it uses a human vision system (HVS) model to calculate the perceptual error between the original image and the halftoned output [29]. Through multiple iterations, the algorithm minimizes the error until it reaches a set threshold or the specified number of iterations, thereby achieving optimal visual results. Qin et al. [33] further enhanced the DBS algorithm by introducing a weighted evaluation function that simultaneously considers perceptual image quality and driving current. The paper proposes a linear model that can accurately predict the driving current of the entire image based on the number of pixel reversals in each row of the image. Use a joint optimization objective function while considering both image quality and driving current. The schematic diagram is shown in Figure 7.

### 3.2 | Color Display Processing Algorithms

Although modern color electronic paper displays have achieved full-color capability, their color gamut remains significantly narrower than that of conventional emissive or transmissive displays such as LCDs, primarily due to inherent structural and material limitations. As a result, images

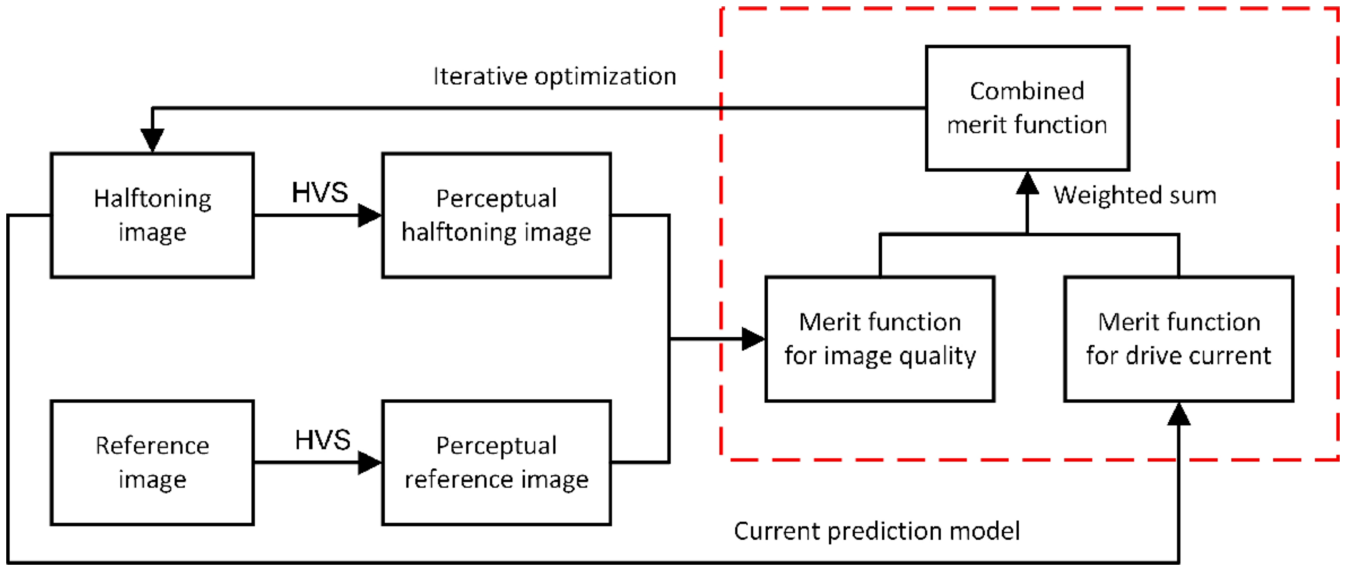


FIGURE 7 | Schematic diagram of the improved DBS algorithm [33].

originally encoded in standard RGB color spaces must undergo gamut mapping to adapt their colors to the limited gamut of the electronic paper before display, followed by halftoning to render the final image. Traditional gamut mapping approaches commonly employ a LUT that directly maps each input RGB triplet to its closest representable color within the device's gamut, offering simplicity and efficiency for hardware implementation.

For instance, in multi-pigment color electronic paper based on CMYW particles, the physical combination of these four pigments can only produce seven primary colors per pixel, leading to a constrained gamut and low contrast. To address this, Kao et al. [34] proposed an advanced color reproduction pipeline. First, a Sigmoidal Gaussian Cusp Knee (SGCK) curve performs nonlinear lightness mapping to enhance mid-tone contrast while aligning cusp points with the device's achievable lightness boundaries, preventing highlight/shadow clipping under strict luminance constraints. Second, a gray axis transformation corrects the inherent neutral-axis misalignment caused by CMY-mixed black pixels, which do not lie on the CIELAB neutral point. Third, hue-angle-specific gamut compression employs a multi-anchor-point strategy: For each hue, anchor points are determined by comparing the cusp lightness values between source and target gamuts. The mapped colors are then rendered via error diffusion halftoning, yielding significantly improved contrast, saturation, and color fidelity compared to naive RGB-to-device mapping.

In contrast, color-filter-based electronic paper suffers from reduced reflectance due to optical absorption in the filters, and cross-talk among color channels as the brightness under one filter can inadvertently influence adjacent color regions. To compensate, Kao et al. [35] developed a nonlinear color model that characterizes the input-output relationship of the display, and used it to pre-enhance input images by boosting brightness, saturation, and contrast before mapping them into the device-specific RGB space, followed by halftoning to ensure smooth visual transitions. The processing flow of two algorithms is shown in Figure 8.

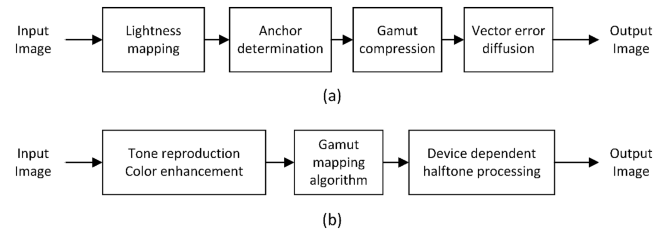
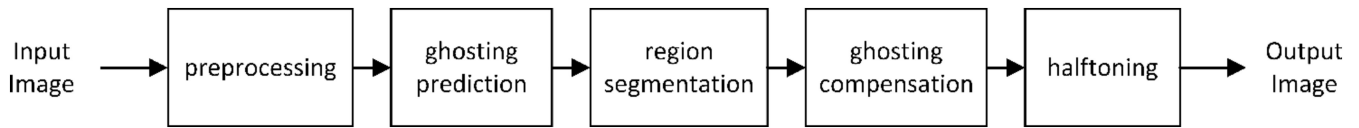


FIGURE 8 | (a) Image processing flow of multi segment color electronic paper [34]; (b) algorithm for electronic paper with color filter [35].

Researchers are now exploring the application of deep learning methods to e-paper image processing. Fang et al. [36] employed a generative adversarial network (GAN) to enable real-time halftone image generation for multi-pigment color EPDs. Tung et al. [37] proposed a DSC-based (Depthwise Separable Convolution) deep learning framework that outperforms conventional methods in color reproduction accuracy, processing speed, and model compactness, making it particularly suitable for CFA e-paper displays.

### 3.3 | Video Display Processing Algorithms

When electronic paper displays (EPDs) are required to render video content, several challenges arise. First, current hardware and underlying technologies are still insufficient to meet the demanding computational requirements for real-time halftoning and image enhancement, resulting in low frame rates. Moreover, rapid switching of the display not only leads to excessive power consumption but also exacerbates ghosting artifacts. Ghosting in EPDs manifests in two primary forms: grayscale ghosting and boundary ghosting. The grayscale ghosting occurs because the driving waveform is unable to completely erase the content of the previous image, resulting in the displayed content having residual image elements. Boundary ghosting, on the other hand, arises from lateral movement of particles caused by horizontal electric fields formed between adjacent electrodes.



**FIGURE 9** | Software-based ghosting effect elimination, region segmentation refers to processing ghosts images into different regions according to the pattern [42].

To address power constraints, Hsu et al. [38] proposed a lossy compression scheme specifically designed for halftoned video on EPDs, which reduces the amount of data that needs to be processed and transmitted, thereby lowering overall system energy consumption. This work represents the first attempt in the field of halftoned video compression for electronic paper.

In parallel, researchers have sought to accelerate processing by integrating halftoning algorithms directly into hardware systems [39]. Kao et al. [40] applied a modified DBS algorithm to video frames, leveraging data from the previous frame to pseudo-randomly reorder pixel processing, thereby minimizing flicker in static regions. Their approach significantly improved video quality compared to conventional error diffusion and the original DBS algorithm. However, due to the high computational complexity of DBS, a novel hardware architecture for parallel video halftoning was subsequently proposed [41], capable of processing four pixels simultaneously with DBS to enhance throughput.

On the software side, Hu et al. [42] developed a model-based method to mitigate ghosting during fast-refresh video playback on EPDs. Their approach predicts the occurrence of both types of ghosting and pre-compensates the source image to counteract the expected artifacts. As illustrated in Figure 9, the algorithm pipeline consists of five stages: preprocessing, ghosting prediction, region segmentation, ghosting compensation, and halftone output. This framework has been demonstrated to be effective for grayscale video display on EPDs.

### 3.4 | Summary of Image Processing Algorithms

E-paper image processing algorithms primarily rely on halftoning and its improved variants to enhance display quality. For color images, additional steps such as gamut conversion and color enhancement are required before halftoning, in order to fully utilize the display characteristics of different types of color e-paper and achieve better visual effects. When implementing video display on e-paper, hardware power consumption and computational limitations must also be taken into account. Although image processing algorithms have already led to significant improvements in display quality, achieving richer full-color and video displays will require the development of more advanced EPD devices to make greater breakthroughs.

## 4 | E-Paper Driving Waveforms

As previously mentioned, the charged particles in electrophoretic e-paper move under the influence of an electric field, thereby forming the displayed image. The operation of EPDs

hinges on the physical migration of charged pigment particles within a viscous dielectric fluid under an applied electric field—a mechanism fundamentally distinct from conventional emissive or transmissive displays. This particle-based switching introduces several interrelated physical constraints that critically shape driving waveform design. First, strict *DC balance* must be maintained over each drive cycle to prevent electrochemical degradation; even minor net DC offsets can reduce device lifetime by more than 50% after 10,000 cycles [43]. Second, the *slow switching dynamics*, typically ranging from 200 ms to 1 s, arise from high fluid viscosity (5–50 cP) and Stokes drag, rendering EPDs inherently incompatible with standard video frame rates (16–30 ms) [14]. Third, *history-dependent ghosting* occurs when particles fail to fully settle, leaving residual optical states that will cause luminance errors in subsequent frames [44]. Additionally, *pixel cross-talk*—induced by lateral electric fields from voltage differences between adjacent pixels—leads to horizontal particle drift and edge blurring over 2–3 pixel widths [45]. Finally, *temperature sensitivity* significantly modulates performance: A 20°C drop can double fluid viscosity, slowing response times by a factor of 2–3 [46]. Collectively, these challenges necessitate driving waveforms that actively manage particle trajectories rather than merely encode image data, transforming waveform design into a problem of electrokinetic control.

### 4.1 | Driving Waveforms for Black-and-White E-Paper

When displaying images with a small number of grayscale levels, complex waveforms are not required, and point-to-point driving waveforms can be used instead. For example, Amundson et al. [18] proposed different grayscale transition methods for four-level black-and-white e-paper. As shown in Figure 10, direct transitions are the fastest but result in the most noticeable grayscale differences. When multi-level grayscale display is required, the number of possible waveform combinations becomes excessive. Therefore, a typical driving waveform for black-and-white electrophoretic e-paper, as shown in Figure 11, is used. It includes three stages [47]: the erase stage, the activation stage, and the write-in stage. The erase stage clears the original image by driving all pixels to either a fully black or fully white state. The activation stage applies a uniform voltage to standardize the distribution of charged particles within the microcapsules. The write-in stage applies different waveforms to different pixel regions based on the image data to render the desired grayscale variations. Typically, the driving voltage ranges between  $\pm 15V$ .

Building upon this structure, optimizing the erase and activation stages can shorten the overall driving time and reduce image ghosting from previous frames. Kao et al. [48] took into account the response delay of electrophoretic e-paper, analyzing how

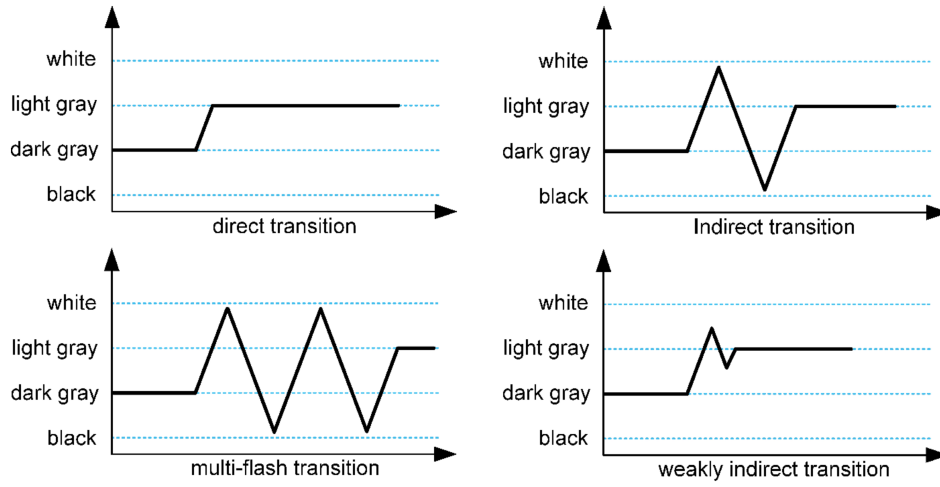


FIGURE 10 | Four grayscale driving waveform [18].

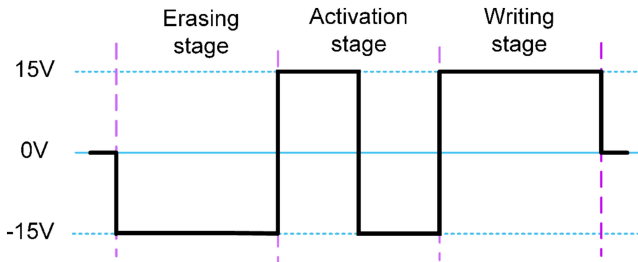


FIGURE 11 | Three-stage driving waveform.

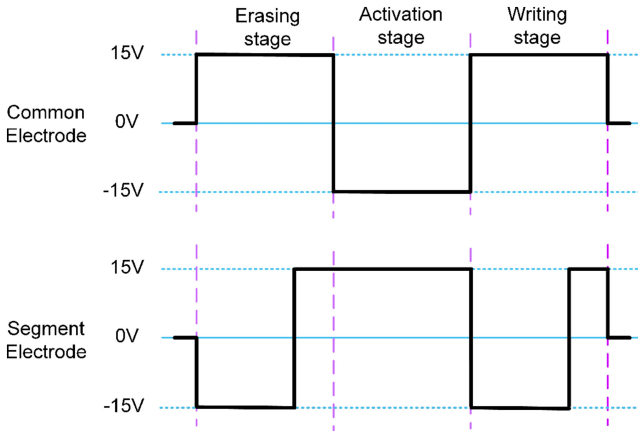


FIGURE 12 | Optimized driving waveform based on response delay [48].

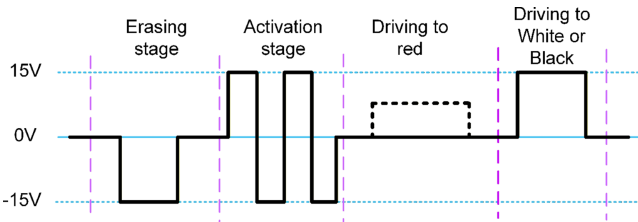


FIGURE 13 | Three-color electronic paper driving waveform [49].

image retention time affects particle movement and the viscosity of the suspension fluid. Based on this, they determined appropriate

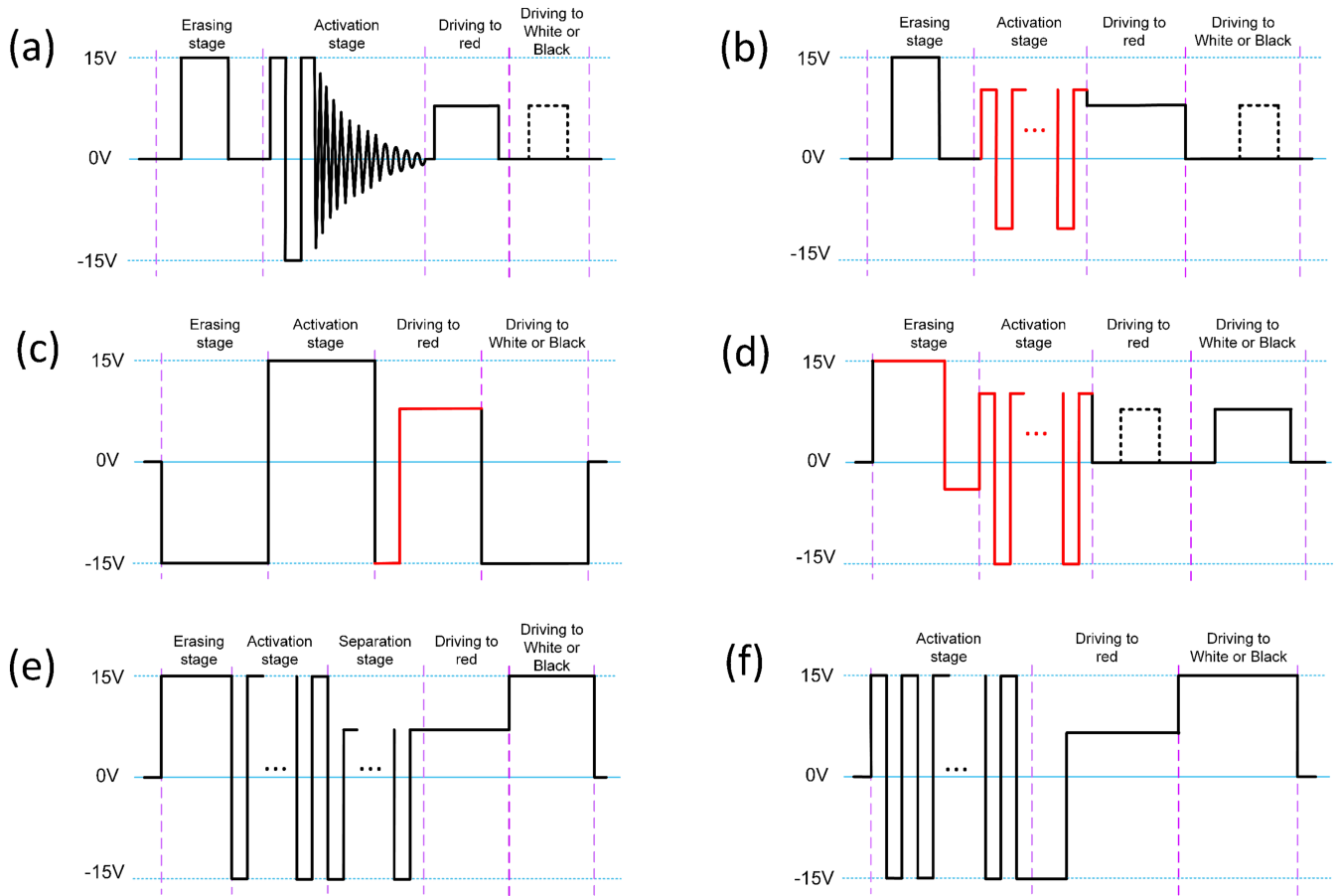
timing compensation parameters to design pulse durations for the erase stage. They also reduced the number of states in the activation stage depending on how much erasure had occurred. These optimizations shortened the total driving time and reduced flicker. Figure 12 shows the improved waveform.

To improve image display quality, further optimization of the write-in stage is also necessary. Shen et al. [45] proposed an optimized waveform design to reduce edge ghosting caused by non-uniform electric field distribution. Their method involved adjusting the voltage sequence to avoid large voltage differences between adjacent pixels, thus minimizing the influence of lateral electric fields on particle movement. They also applied synchronous vertical drive sequences to reduce grayscale inconsistencies by encouraging vertical particle blending.

## 4.2 | Driving Waveforms for Tri-Color E-Paper

In black-white-red tri-color electrophoretic e-paper, the white particles are negatively charged, while the black and red particles are positively charged. Due to differences in surface charge density (Zeta potential), particle size, and mass, black and red particles exhibit distinct electrophoretic mobilities and threshold voltages. Specifically, black particles possess a higher Zeta potential compared to red particles, resulting in faster migration speed under the same electric field but requiring a higher threshold voltage for activation. By applying appropriately selected high and low positive voltages, black and red particles can be driven independently, enabling the display of three primary color states (white, black, and red) as well as mixed-color states including black-red. As shown in Figure 13, Kao et al. proposed a four-stage driving waveform for black-white-red e-paper [49], in which the original write-in stage is divided into a red-particle driving stage and a white/black-particle driving stage.

Building upon this four-stage structure, researchers have optimized each stage to improve display quality. Yi et al. [50] proposed using a damped oscillation waveform during the activation stage to separate red and black particles. As the duration of the oscillation increases, the red and black particles gradually separate, but excessive duration causes them to recombine. Figure 14a illustrates the



**FIGURE 14** | Improved three color electronic paper driving waveform: (a) use damping oscillation during the activation phase [50]; (b) optimize the oscillation frequency and time during the activation phase [51]; (c) adjust the duration of the activation phase [52]; (d) adjust the erase phase voltage and time to match the activation phase cycle [53]; (e) add separation stage [54]; (f) delete the erase phase [55].

third stage as a damped oscillation waveform. The particle state after the activation stage becomes the reference grayscale for the write-in stage; if the reference position is close to the target grayscale position, the required driving time in the write-in stage can be reduced. Zhang et al. [51] analyzed how the oscillation period and cycle count in the activation stage affect particle behavior. By adjusting these parameters, they reduced the distance between the particles and the common electrode, shortening the response time of black and white particles in tri-color e-paper. Figure 14b shows the optimized waveform. Zeng et al. [52] shortened the response time of red particles by reducing the distance between red particles and the electrode through adjusting the activation stage duration—see Figure 14c. Adjusting the erase stage has also proven effective in reducing ghosting. Jiang et al. [53] found that setting appropriate voltage amplitude and duration ratios in the erase stage minimized ghosting from red particles. Figure 14d shows the corresponding driving waveform.

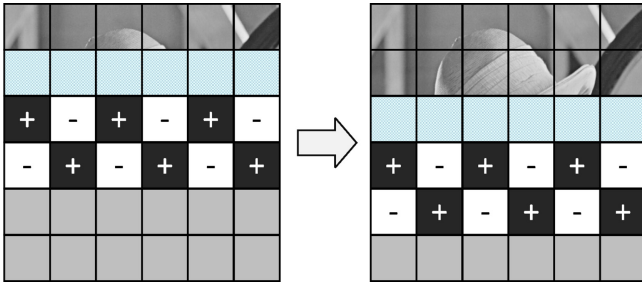
Beyond optimizing the four defined stages, researchers have also modified the structure of the waveform itself to further enhance image quality. Liu et al. [54] inserted an additional separation stage before the write-in phase to separate red and black particles. This reduced pre-write color mixing and improved red saturation. The waveform, shown in Figure 14e, includes five stages: erase, activation, separation, red driving, and black-white driving. Zhang et al. [55] took a different

approach by removing the erase stage entirely. They proposed a three-stage waveform—activation, red particle driving, and black or white particle driving—as shown in Figure 14f. By adjusting the oscillation period and duration in the activation stage and optimizing the driving voltage for red particles, they reduced the response times for all three types of particles: black, white, and red.

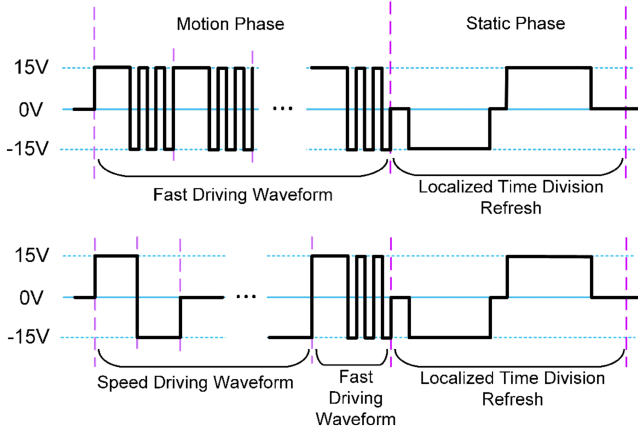
### 4.3 | Driving Waveforms for Video Display on E-Paper

Compared to the driving waveforms used for still image display, those for video display on e-paper have much shorter available durations. As a result, there is insufficient time to fully erase the previous image, making residual ghosting more likely. Researchers must optimize driving waveforms to reduce ghosting while also improving the refresh rate of e-paper.

To improve refresh speed, it is essential to reduce the driving time per video frame. By comparing the current frame data with the previous frame data and directly generating EPD drive signals [56], memory usage and driving time can be minimized. Wang et al. [57] proposed a method that selects a shorter grayscale transition path based on the grayscale difference between consecutive video frames and designed a set of



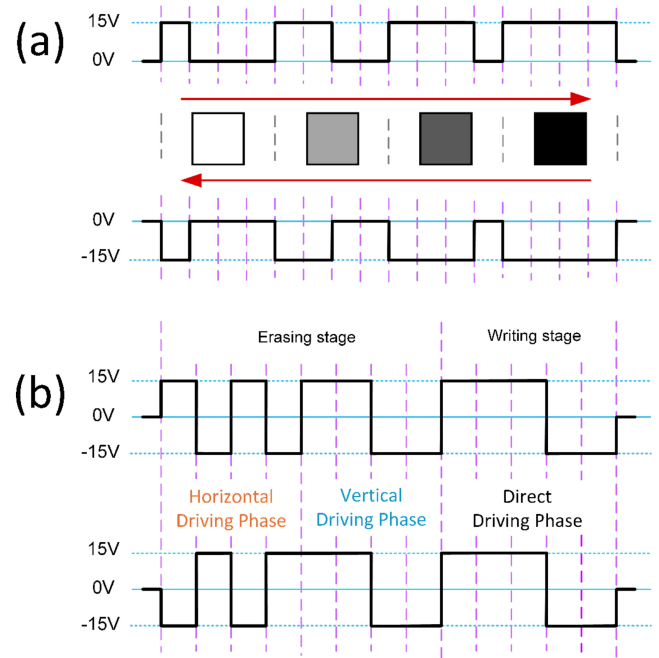
**FIGURE 15** | Perform erase and activate phases in different regions; LTDR reduces perceived flicker by > 60% [58].



**FIGURE 16** | Dynamic display waveform combined with LTDR [58].

waveforms with shorter drive times. Li et al. [58] introduced a dynamic multi-waveform invocation method along with a local time-division refresh (LTDR) technique to address issues of refresh speed and flickering. The traditional three-stage driving waveform is compressed and modified to obtain the LTDR driving waveform used for static image refresh. The erase and activate phases are interleaved in different regions to reduce flicker during refresh, as shown in Figure 15. By combining 16 cycles of fast driving waveforms and 4 cycles of speed driving waveforms with LTDR, two driving methods for video display were obtained as shown in Figure 16, dynamic and static regions are distinguished by frame-to-frame pixel differences.

To address edge ghosting caused by adjacent pixels, Kao et al. [59] enhanced the waveform by briefly driving pixels to intermediate gray states during the activation stage and then immediately returning them to their original black or white state. This reduced lateral disturbances. Wang et al. [60] targeted both grayscale and edge ghosting in video playback by designing two driving methods: Multi-frame pulse width modulation (MFPWM) adjusts the pulse width incrementally over several frames to refine particle positioning, as shown in Figure 17a, unlike traditional PWM, which determines grayscale within a single frame (Figure 10). Time-sharing longitudinal and transverse oscillation (TSLTO), shown in Figure 17b, uses lateral and vertical electric fields before the write-in stage to thoroughly mix particles. These techniques effectively reduce both types of ghosting and significantly improve video performance on e-paper displays.



**FIGURE 17** | Reduce residual image driving waveform [60]: (a) MFPWM; (b) TSLTO.

#### 4.4 | Summary of E-Paper Driving Waveforms

The optimization design of driving waveforms for EPD primarily focuses on the following three objectives: 1. Achieving more and more precise grayscale levels; 2. Reducing ghosting from residual particle states after image updates; 3. Shortening the required driving time. At present, most designs use high-frequency oscillating voltages to erase previous images and to establish a uniformly distributed reference grayscale. During the write-in stage, waveform design increasingly considers the physical properties of microcapsules to improve image display quality. Given the physical limitations of EPD, designers must strike a balance between driving time and display quality according to specific application requirements.

### 5 | Development Trends

After decades of development, EPD technology has made significant progress, but some advanced innovations still face challenges in mass production and commercialization. Bridging this gap requires advancements in several key areas:

1. Enhancing grayscale and color gamut: To address color limitations, better gamut mapping and enhancement algorithms are needed to improve color fidelity and visual quality. Color correction based on deep learning is a feasible solution.
2. Optimizing dynamic display: As EPD expands into dynamic applications like video playback and scrolling, algorithms and waveforms must minimize flicker and ghosting to enhance clarity and smoothness.
3. Incorporating visual perception: EPD performance is affected by ambient lighting, so image processing should

account for human visual perception. Enhancing grayscale, color, and detail based on how the eye perceives them can improve visual comfort and perceived quality.

4. Personalized waveform design: Tailoring waveforms by grayscale level, content type (text, image, video), and refresh frequency can optimize visual quality and energy efficiency. Customizing waveforms based on particle properties, fluid composition, and device structure can further boost performance.
5. Co-optimization with new materials and structures: As new EPD materials and architectures emerge—such as advanced particles, fluids, and color technologies—image algorithms and waveforms should be co-optimized with these innovations to improve contrast, refresh rates, and color gamut comprehensively.

## 6 | Conclusion

The current prospects for e-paper technology are promising, and its applications have expanded from initial use in e-book readers to fields such as retail, transportation, and education. Although e-paper has made progress in color display, it still lags behind other display technologies in certain aspects. For example, full-color video playback faces challenges such as color performance and refresh rates. Additionally, the high manufacturing cost of e-paper limits its widespread adoption. While electrophoretic e-paper still has several shortcomings to overcome, its low power consumption and eye-friendly features make it an indispensable part of the display technology landscape.

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### Conflicts of Interest

The authors declare no conflicts of interest.

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